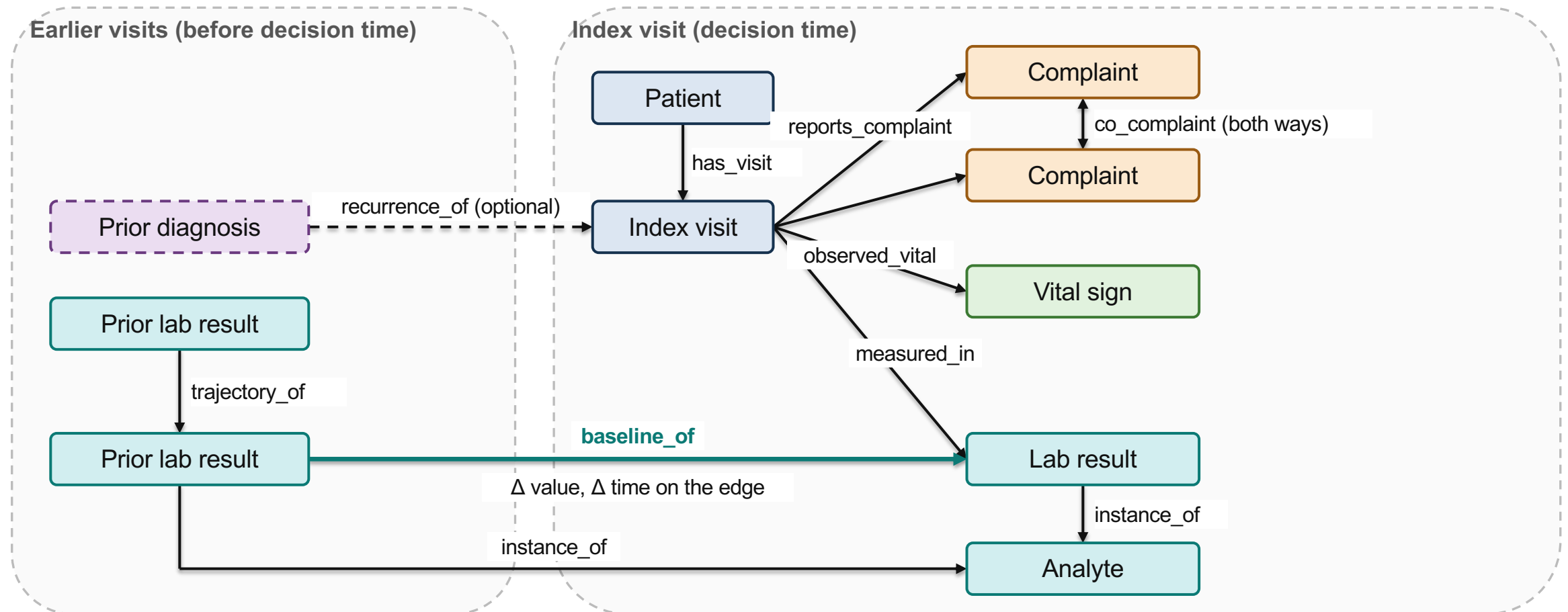


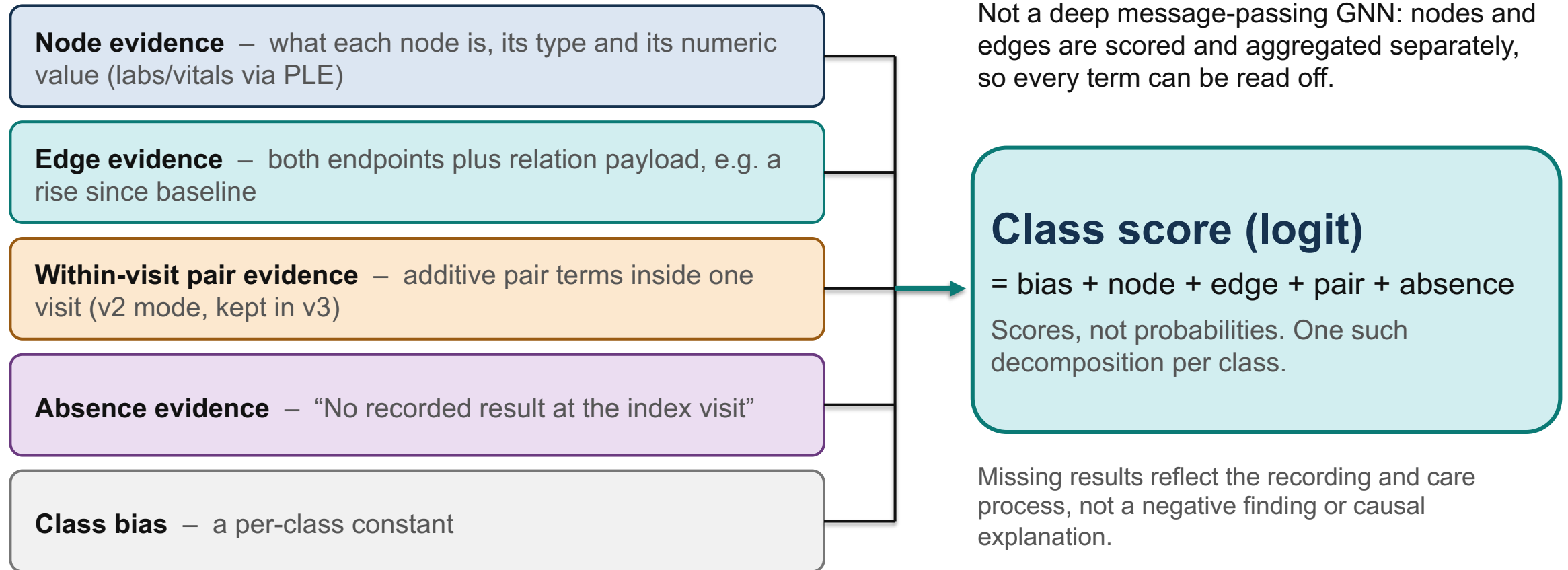
A shared clinical graph, not a flat patient record

One typed graph per decision-time visit: patient, visits, complaints, labs and vitals, with numeric change between earlier and index results kept on the edge. Saved as a reusable project asset.



CEI-GNN: a diagnosis score you can decompose

Each class score is an exact sum of signed evidence terms: positive supports the class, negative opposes it.



PLE = piecewise-linear encoding of numeric values, fitted on training data (K = 4 selected). CEI v3 builds on the additive v2 pair mode; the multiplicative pair interaction was not successful.

Prediction and explanation quality

Primary screen: macro-F1

Method	Macro-F1	Active params
CEI-GNN v3	0.6418	94,380
CEI-GNN v2	0.6420	92,300
ProtGNN	0.6371	399,884
XGBoost	0.6371	N/A

Patient-equal macro-F1; 5,000 screen visits / 3 seeds.

v2 control: 94,380 total, 92,300 active.
XGBoost: not a neural parameter count.

No decisive accuracy gain on the primary screen.

Explanation quality: CEI / ProtGNN (3-seed means)

Explainer	Fidelity- (lower better)	Fidelity+ (higher better)	Sparsity (higher better)
Native	0.105 / N/A	0.554 / N/A	0.700 / N/A
GradExplainer	0.144 / 0.571	0.509 / 0.510	0.513 / 0.388
Integrated Gradients	0.100 / 0.570	0.538 / 0.535	0.521 / 0.289
GNNExplainer	0.398 / 0.576	0.216 / 0.381	0.269 / 0.270
Random	0.484 / 0.578	0.107 / 0.309	0.285 / 0.285

500 screen visits / 498 subjects / 3 seeds. Native ProtGNN attribution unavailable in this implementation.

Grad/IG: lower fidelity- and higher sparsity; fidelity+ gains not established.
GNNExplainer fidelity+ favors ProtGNN.

Faithfulness to each model, not clinical validity. Held-out test not used.

Fidelity-: probability drop when only important nodes are kept. Fidelity+: drop when important nodes are masked. Sparsity: fraction excluded at 90% importance mass.